

Emotion / imagination
Reason / logic

Srinivasa Ramanujan

— C.P. Snow

Introduction

Srinivasa Ramanujan was born in Erode in 1887. He had his schooling at Kumbakonam, a small temple town in South India. His obsession with mathematics was such that at age 13 he mastered trigonometry and made his own theorems. But since he was not good in other subjects he could not complete his degree. He worked in Chennai Port Trust, where, after completing his office work, he would pursue his own research in mathematics.

His life underwent a sea change when his genius was recognized by the great mathematician Hardy and Ramanujan sailed to Cambridge in 1914. After the completion of his research he was made the fellow of the Royal society in 1918. He also became the first Indian Fellow of Trinity College, Cambridge. He returned to India in 1919 but his ill-health took its toll and he died the same year when he was hardly 33.

About the text

This extract is from the book *Variety of Men* by C.P. Snow. This passage on Ramanujan, is a part of the biographical sketch of the mathematician Hardy and it tells us how Hardy 'discovered' Ramanujan. Ramanujan had sent to Hardy some theorems without proofs and had asked for his opinion. Hardy at first wondered if it was the work of a fraud, but was later convinced that Ramanujan was really a genius. Hardy arranged to bring Ramanujan to Cambridge in 1914. Ramanujan worked hard in Cambridge for five years. Hardy and Ramanujan together produced five papers of the highest class. This extract brings out the genius as well as the fine human qualities in both Hardy and Ramanujan.

Reading

About his discovery of Ramanujan, Hardy showed no secrecy at all. It was, he wrote, the one romantic incident in his life: anyway, it is an admirable story, and one which showers credit on nearly everyone (with two exceptions) in it. One morning, early in 1913, he found, among the letters on his breakfast table, a large untidy envelope decorated with Indian stamps. When he opened it, he found sheets of paper by no means clean, on which, in a non-English script, were line after line of symbols. Hardy glanced at them without enthusiasm. He was by this time, at the age of thirty-six, a world famous mathematician; and world famous mathematicians, he had discovered, are unusually exposed to cranks. He was accustomed to receiving manuscripts from strangers, proving the prophetic wisdom of the Great Pyramid, the revelations of the Elders of Zion, or the cryptograms that Bacon had inserted in the plays of the so-called Shakespeare.

So Hardy felt, more than anything, bored. He glanced at the letter, written in halting English, signed by an unknown Indian, asking him to give an opinion of these mathematical discoveries. The script appeared to consist of theorems, most of them, wild or fantastic looking, one or two already well-known, laid out as though they were original. There were no proofs of any kind. Hardy was not only bored, but irritated. It seemed like a curious kind of fraud. He put the manuscript aside, and went on with his day's routine. Since his routine did not vary throughout his life, it is possible to reconstruct it. First he read the *Times* over

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his breakfast. This happened in January, and if there were any Australian cricket scores, he would start with them, studied with clarity and intense attention.

Then, from about nine to one, unless he was giving a lecture, he worked at his own mathematics. Four hours' creative work a day is about the limit for a mathematician, he used to say. Lunch, a light meal, in hall. After lunch he loped off for a real game of tennis in the university court. (If it had been summer, he would have walked down to Fenner's to watch cricket.) In the late afternoon, a stroll back to his rooms. That particular day, though, while the time-table wasn't altered, internally things were not going according to plan. At the back of his mind, getting in the way of his complete pleasure in his game, the Indian manuscript nagged away. Wild theorems. Theorems such as he had never seen before, nor imagined. A fraud of genius?

A question was forming itself in his mind. As it was Hardy's mind, the question was forming itself with epigrammatic clarity: is a fraud of genius more probable than an unknown mathematician of genius?

Clearly the answer was no. Back in his rooms in Trinity, he had another look at the script. He sent word to Littlewood (probably by messenger, certainly not by telephone, for which, like all mechanical contrivances including fountain pens, he had a deep distrust) that they must have a discussion after all.

When the meal was over, there may have been a slight delay. Hardy liked a glass of wine, but, despite the glorious vistas of "Alan St. Aubyn" which had fired his youthful imagination, he found he did not really enjoy lingering in the combination-room over port and walnuts. Littlewood, a good deal more pleasure-loving, did. So there may have been a delay. Anyway, by nine o'clock or so they were in one of Hardy's rooms, with the manuscript stretched out in front of them.

That is an occasion at which one would have liked to be present. Hardy with his combination of remorseless clarity and intellectual dash (he was very English, but in argument he showed the characteristics that Latin minds have often assumed to be their own). Littlewood, imaginative, powerful, humorous, apparently it did not take them long. Before midnight they knew, and knew for certain. The writer of these manuscripts was a man of genius. That was as much as they could judge, that night. It was only later that Hardy decided that Ramanujan was, in terms of *natural* mathematical genius, in the class of Gauss and Euler; but that he could not expect, because of the defects of his education and because he had come on the scene too late in the line of mathematical history, to make a contribution on the same scale.

It all sounds easy, the kind of judgment great mathematicians should have been able to make. But I mentioned there were two persons who do not come out of the story with credit. Out of chivalry Hardy concealed this in all that he said or wrote about Ramanujan. The two people concerned have now been dead, however, for many years, and it is time to tell the truth. It is simple. Hardy was not the first eminent mathematician to be sent the Ramanujan manuscripts. There had been two before him, both English, both of the highest professional standing. They had each returned the manuscript without comment. I don't think history relates what they said, if anything, when Ramanujan became famous. Anyone who has been sent unsolicited material will have a sneaking sympathy with them.

Anyway, the following day Hardy went into action. Ramanujan must be brought to England, Hardy decided. Money was not a major problem. Trinity has usually been good at supporting unorthodox talent (the college did the same for Kapitsa a few years later). Once Hardy was

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determined, no human agency could have stopped Ramanujan, but they needed a certain amount of help from a superhuman one.

Ramanujan turned out to be a poor clerk in Madras, living with his wife on twenty pounds a year. But he was also a Brahmin, usually strict about his religious observances, with a mother who was even stricter. It seemed impossible that he could break the ban and cross the waters. Fortunately his mother had the highest respect for the goddess of Namakkal. One morning Ramanujan's mother made a startling announcement. She had a dream the previous night in which she saw her son seated in a big hall among a group of Europeans and the goddess of Namakkal had commanded her not to stand in the way of her son fulfilling his life's purpose. This, say Ramanujan's Indian biographers, was a very agreeable surprise to all concerned.

In 1914 Ramanujan arrived in England. So far as Hardy could detect (though in this respect I should not trust his insight far) Ramanujan, despite the difficulties of breaking the caste laws, did not believe much in theological doctrine, except for a vague pantheistic benevolence, any more than Hardy did himself. But he did certainly believe in ritual. When Trinity put him up in college—within four years he became a Fellow—there was no "Alan St. Aubyn" self-indulgence for him at all. Hardy used to find him ritually changing into his pyjamas, cooking vegetables rather miserably in a frying-pan in his own room.

Their association was a strangely touching one. Hardy did not forget that he was in the presence of genius; but genius that was, even in mathematics, almost untrained. Ramanujan had not been able to enter Madras University because he could not matriculate in English. Accordingly to Hardy's report, he was always amiable and good-natured, but no doubt he sometimes found Hardy's conversation outside mathematics more than a little baffling. He seemed to have listened with a patient smile on his good, friendly, homely face. Even inside mathematics they had to come to terms with the difference in their education. Ramanujan was self-taught; he knew nothing of the modern rigour; in a sense he didn't know what a proof was. In an uncharacteristically sentimental moment, Hardy once wrote that if he had been better educated, he would have been less Ramanujan. Coming back to his ironic senses, Hardy later corrected himself and said that the statement was nonsense. If Ramanujan had been better educated, he would have been even more wonderful than he was. In fact, Hardy was obliged to teach him some formal mathematics as though Ramanujan had been a scholarship candidate at Winchester. Hardy said that this was the most singular experience of his life; what did modern mathematics look like to someone who had the deepest insight, but who had literally never heard of most of it?

Anyway, they produced together five papers of the highest class, in which Hardy showed supreme originality of his own (more is known of the details of this collaboration than of the Hardy-Littlewood one). Generosity and imagination were, for once, rewarded in full.

This is a story of human virtue. Once people had started behaving well, they went on behaving better. It is good to remember that England gave Ramanujan such honours as were possible. The Royal Society elected him a Fellow at the age of thirty (which, even for a mathematician, is very young). Trinity also elected him a Fellow in the same year. He was the first Indian to be given either of these distinctions. He was amiably grateful. But he soon became ill.

Hardy used to visit him, as he lay dying in hospital at Putney. It was on one of those visits that there happened the incident of the taxi-cab number. Hardy had gone out to Putney by taxi, as usual his chosen method of conveyance. He went into the room where Ramanujan was lying. Hardy, always clumsy about introducing a conversation, said, probably without a greeting and certainly as his first remark: "The number of my taxi-cab was 1729. It seems to me rather a dull

number." To which Ramanujan replied: "No, Hardy! No, Hardy! It is a very interesting number. *It is the smallest number expressible as the sum of two cubes in two different ways.*"

This is the exchange as Hardy recorded it. It must be substantially accurate. He was the most honest of men and further, no one could possibly have invented it.

It was difficult, in wartime, to move Ramanujan to a kinder climate. He died of tuberculosis, back in Madras, two years after the war. As Hardy wrote in the Apology, in his roll-call of mathematicians: "Galois died at twenty-one, Abel at twenty-seven, Ramanujan at thirty-three, Riemann at forty. I do not know an instance of a major mathematical advance initiated by a man past fifty".